

RAMESH R

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Professional Summary

Experienced engineering professional with over 2 years of comprehensive experience in software development, specializing in the Automotive domain with a focus on Telematics. Proficient in all aspects of the software development lifecycle, development, debugging, testing, and maintenance. Extensive expertise in V2X stack, Vanetza stack, C-V2X, and C-ITS, with a proven track record of successfully implementing and testing V2X features. Skilled in C/C++ programming and proficient in corresponding libraries such as Boost and Crypto. Possesses a strong foundation in Object-Oriented Design and Analysis, Data Structures and Algorithms, Sockets, SOMEIP, SUMO, and Protobuf.

Experience

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August 2022 – curr

Senior Software Engineer, Telematics V2X

Bangalore, Karnataka

- Experienced software developer with expertise in C++ programming and a focus on developing modules for Cooperative Intelligent Transport Systems (C-ITS).
- A detail-oriented team member with strong problem-solving skills, dedicated to achieving project success..
- Proven ability to contribute effectively to the software development lifecycle, ensuring the delivery of high-quality solutions.
- Utilized Protocol Buffers for efficient serialization and deserialization of data structures.
- Designed message formats and defined service interfaces using Protobuf's language-agnostic protocol.
- Leveraged Visual Studio Code as the primary Integrated Development Environment (IDE) for V2X stack development.
- Worked extensively in Ubuntu and Linux environments for V2X stack development and deployment
- Implemented SOMEIP protocol support within the V2X stack for inter-process communication.
- Handled memory management and resource allocation effectively to ensure stability and reliability.

Technical Skills

Programming Languages: C , C++

Domain specific skills: Linux / C++ , Design patterns, UML , IPC Mechanisms (message queue, shared memory, SOMEIP) , Unit Testing / Gtest - Telematics/ V2X/ IVI (using C/ C++) , Boost C++ Libraries , G-Test , Git , Jira , Linux , Static Analysis, TCP/IP , UDP , SOMEIP, UDS

Additional Skills: The Mathworks-Matlab, Ubuntu , Virtual Machine , Visual studio

Tools: SUMO-GUI, lcov, gcov, putty, CentOS, CMake, ubuntu, windows, netedit, MakeFiles

Projects

SELinux - based File Access Control and Domain Isolation!

- Designed and developed custom SELinux policies to manage file access control between different security domains
- Configured domain transition rules where one domain was granted read - only access and another domain had read - write permissions.
- Ensured that unauthorized write attempts from the read - only domain were effectively blocked, maintaining strict access controls and preventing unauthorized file modifications.
- Experience in managing security through SELinux, emphasizing your understanding of domain transitions and access restrictions.

Education

University college of engineering, BIT campus

Bachelor of Technology in Electrical and Electronics Engineering

Aug 2018 – May 2022

Thiruchurapalli, Tamilnadu

Sannathi Higher Secondary School

Class XII

Aug 2017 – May 2018

Thiruvannamalai, Tamilnadu

Sannathi Higher Secondary School

Class X

June 2015 – March 2016

Thiruvannamalai, Tamilnadu

Relevant Coursework

- Object Oriented Programming with C++
- Programming with C
- Operating Systems
- Networking Protocols